

Scalable ML Assignment

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1 Log Mining and Analysis

Table 1: Total requests and unique hostnames by UK sector.

Sector	Total Requests	Unique Hosts
Academic	25009	1022
Company	22225	1116
Government	195	15

For the `.ac.uk` hosts, the institution domain is extracted by first splitting each hostname using the full stop delimiter. For example, the Sheffield hostname, `www.shef.ac.uk`, is split into `["www", "shef", "ac", "uk"]`. The final three labels are concatenated together, resulting in `shef.ac.uk`, which represents the institution domain. Different hosts from the same institution (`cs.shef.ac.uk`, `www.shef.ac.uk`, etc.) are grouped together within the Sheffield domain.

Listing 1 shows the Spark implementation of the method described above.

Listing 1: Institution domain extraction and aggregation.

```
academic_hosts = logs.filter(col("host").endsWith(".ac.uk"))

split_host = split(col("host"), "\\.")

academic_domains = academic_hosts.select(
    col("host"),
    concat_ws(".",
        element_at(split_host, -3),
        element_at(split_host, -2),
        element_at(split_host, -1)
    ).alias("institution")
)

institution_counts = academic_domains.groupBy("institution") \
    .count() \
    .withColumnRenamed("count", "total_requests")
```

The total number of requests from the University of Sheffield domain is 623. The seven institutions with more requests than Sheffield, and their respective total requests, are shown below:

Table 2: UK academic institutions with more requests than the University of Sheffield.

Institution	Total Requests
<code>hensa.ac.uk</code>	4257
<code>rl.ac.uk</code>	1158
<code>ucl.ac.uk</code>	1036
<code>man.ac.uk</code>	921
<code>ic.ac.uk</code>	851
<code>soton.ac.uk</code>	808
<code>bham.ac.uk</code>	629